Auteur: Seda Jusjajeva
Studierichting: Informatica
Oefeningenreeks 5F nr. 1.1

$r = -\cos\theta$

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$$S = 2 \cdot \frac{1}{2} \int_{\frac{\pi}{2}}^{\pi} \cos^2\theta d\theta$$

$$\int \cos^2\theta d\theta = \int \frac{1+\cos 2\theta}{2} d\theta = \frac{1}{2}\int d\theta + \frac{1}{2}\int \cos 2\theta d\theta = \frac{\theta}{2} + \frac{\sin 2\theta}{4} + c$$

$$\Rightarrow S = \left[\frac{\theta}{2} + \frac{\sin(2\theta)}{4}\right]_{\frac{\pi}{2}}^{\pi} = \frac{\pi}{2} + \frac{\sin(2\pi)}{2} - \left(\frac{\pi}{4} + \frac{\sin(\pi)}{2}\right) = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$$

References